21st National High School Physics Competition Final Problems

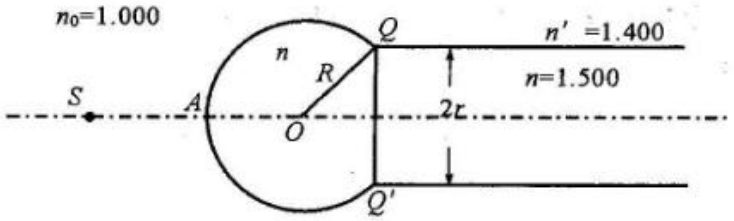
1. (20 points)

Highly transparent fibre core: $n=1.500$, radius r; cladding index $n'=1.400$. Semiconductor laser S, on the axis extension, emits a beam of half-angle 30°. To admit all the light, a spherical cap QAQ′ of radius R, of the core material, is sintered at its end. The figure omits the cladding. S is a point source; the fibre is in air, with index n_0 taken as 1.000.

1. For the whole beam to reach the cap, what condition must distance x from source S to A satisfy?

2. For $R=1.8r$, S–A separation R, axis angle α: for α at $\alpha_1=30°$、$\alpha_2=25°$ and $\alpha_3=20°$, can each of the three rays propagate in the fibre by total internal reflection?

一、有一光导纤维，光芯为折射率 $n=1.500$ 的透明度极好的介质，其截面半径为 r；光芯外面包层的折射率 $n'=1.400$. 有一半导体激光器 S，位于光纤轴线的延长线上，发出半角宽为 30° 的光束. 为便于使此光束全部进入光纤，在光纤端面处烧结了一个其材料与光芯相同、半径为 R 的球冠 QAQ'，端面附近的结构如图所示（包层未画出）. S 可看作点光源，光纤放在空气中，空气的折射率 n_0 按 1.000 计算.



1. 若要半导体激光器发出的光能够全部射到球冠上，则光源 S 离 A 的距离 x 应满足什么条件？

2. 如果 $R=1.8r$，光源 S 与 A 点的距离为 R，入射线与轴的夹角用 α 表示，则 α 角分别为 $\alpha_1=30°$、$\alpha_2=25°$ 和 $\alpha_3=20°$ 的三根光线能否经过全反射在光纤中传播？

2. (20 points)

Use relativistic energy and momentum to analyse and justify the following.

1. A photon colliding with a free electron at rest in vacuum cannot have its energy fully absorbed.

2. A photon on a metal can be fully absorbed, ejecting an electron by the photoelectric effect.

3. (25 points)

Shown wooden wedge OABC ($M=30.0$kg) rests on level ground; AB has slope $\beta=60°$, and BC slope $\alpha=45°$. A very short smooth arc joins AB, BC at B. Release a point block, $m=4.00$kg, from rest at A, height $h_1=2.80$m above ground. It slides without friction on the wedge. Maximum static and sliding ground friction have coefficients $\mu=6.00\times10^{-2}$. The height of B above the table is $h_2=2.00$m. Ignore both speed changes at B. Find work done on the wedge from A to the bottom C by the sliding block. Take $g=10.0\text{m/s}^2$.